Premium RAG Chatbot Documentation Bundle

A polished documentation pack for an internal RAG chatbot over company documents, including the PRD, technical and design stack, and implementation workflow. The recommendations emphasize retrieval quality, hybrid search, permissions, evaluation, and premium product design grounded in current RAG architecture guidance [cite:18][cite:19].

PRD | Tech Stack | Design Stack | Workflow

Product Requirements

This product is a secure internal AI assistant that retrieves approved company knowledge and answers with grounded citations rather than acting as a generic chat layer [cite:18]. The core value is faster internal retrieval, better consistency, permission-aware answers, and a path from answer to operational action [cite:18].

The product should begin with a narrow, high-value pilot use case because enterprise RAG systems become trustworthy through staged rollout and evaluation rather than broad first-release scope [cite:18].

Core goals

- Reduce information search time across fragmented company sources [cite:18].
- Improve answer consistency with citation-backed outputs [cite:18].
- Respect role-based and source-level permissions before generation [cite:18].
- Support workflow follow-through after answers are generated [cite:18].

Critical requirements

- Structured ingestion with business metadata is mandatory because retrieval quality starts at the ingestion layer [cite:18].
- Hybrid retrieval plus reranking is required because semantic search alone often misses exact identifiers and literal business terms [cite:18].
- Evaluation must measure precision, recall, faithfulness, relevance, citations, latency, and escalation behavior before expansion [cite:18].

Tech and Design Stack

LlamaIndex is the strongest default for the retrieval core because it is RAG-first and generally ships document-heavy Q&A systems faster with stronger defaults than more general frameworks [cite:19][cite:2]. LangChain or LangGraph can be added when the product expands into broader orchestration, tools, or multi-step workflows [cite:19][cite:22].

Layer	Recommendation	Reason
Frontend	Next.js + TypeScript + Tailwind + Framer Motion	Premium component architecture and polished animation.
RAG Core	LlamaIndex	RAG-first defaults and strong document handling [cite:19].
Workflow	LangChain or LangGraph	Useful when the app expands beyond document Q&A [cite:22].
Prototype Vector DB	Chroma	Developer-friendly for prototypes and small-scale production [cite:3].
Production Vector DB	Weaviate, Pinecone, or Qdrant	Better scale and filtering performance than Chroma at larger sizes [cite:17][cite:20].
Evaluation	RAGAS + tracing	Production systems need measurable retrieval and answer quality [cite:18].

The design stack should avoid the default neon “AI” look and instead use warm neutral surfaces, elegant type contrast, subtle motion, and source-first answer layouts. Premium trust signals should be visible in the interface through citations, access labels, source previews, and restrained motion.

Workflow Document

The recommended build sequence is use-case definition, source mapping, ingestion, chunking and indexing, hybrid retrieval, grounded generation, evaluation, and then workflow integration, because reliable enterprise RAG is built in layers [cite:18]. This keeps product complexity aligned with evidence and reduces the risk of flashy but untrustworthy automation [cite:18].

Sprint	Focus	Main output
1	Discovery	Use case, source map, metadata model, pilot query set
2	Ingestion	Parsing, chunking rules, indexing scripts
3	Retrieval	Hybrid search, filters, reranker, baseline QA
4	Premium UI	Chat surface, citations, source drawer, motion system
5	Governance	Access control, audits, evaluation harness
6	Actions	Task/report/CRM handoff features
7	Launch	Polish, pilot rollout, feedback loop

This bundle is designed as a premium starting point for stakeholder sharing and product planning. It combines product framing with concrete implementation guidance grounded in current RAG architecture and framework trade-offs [cite:18][cite:19][cite:20].